

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Telecommunication-BBOU/ELEP/AKAB

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Revision History

Rev No.	Date	Description of Revision
R01	05.07.22	Issued for Review
R02	20.10.22	Re-issued for review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-041

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

CCTV	Closed Circuit Television
EPABX	Electronic Private Automatic Branch Exchange
IEC	International Electro-technical Commission
ISO	International Standardisation Organisation
ITU	International Telecommunication Union
MUX	Multiplexer
MODEM	Modulation-Demodulation
SSPA	Solid State Power Amplifier
LNA	Low Noise Amplifier
SCS	Satellite Communication System
AMC	Annual Maintenance Contract
CAMC	Comprehensive annual maintenance contract
SCADA	Supervisory Control and Data Acquisition System
DCS	Distributed Control System

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

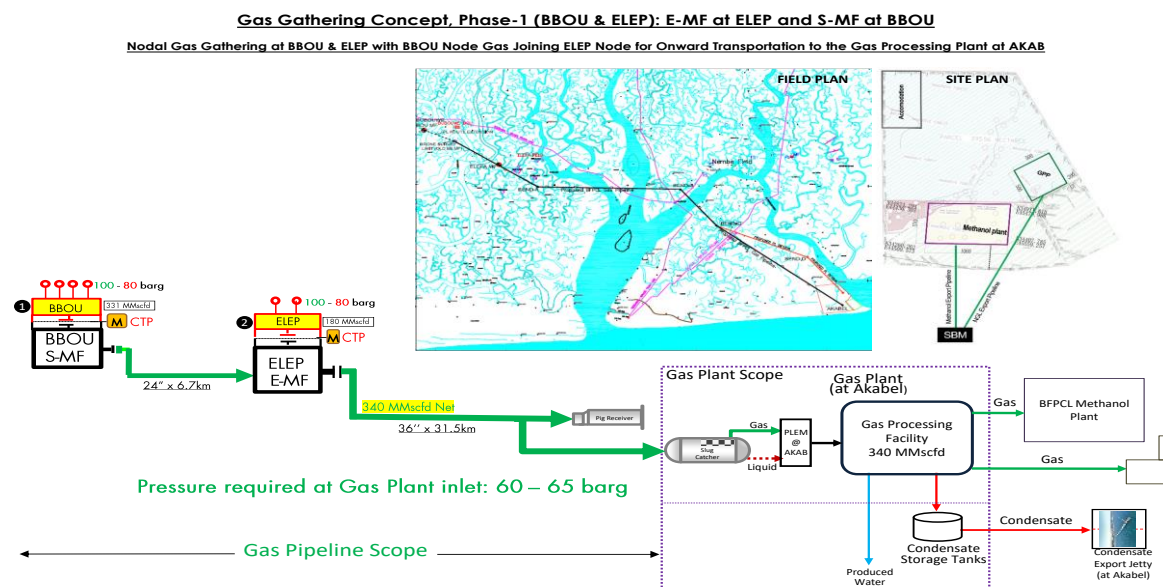


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, software applications, Installation, Inspection, testing, certification and delivery of Satellite Communication systems at BBOU, ELEP and AKAB.

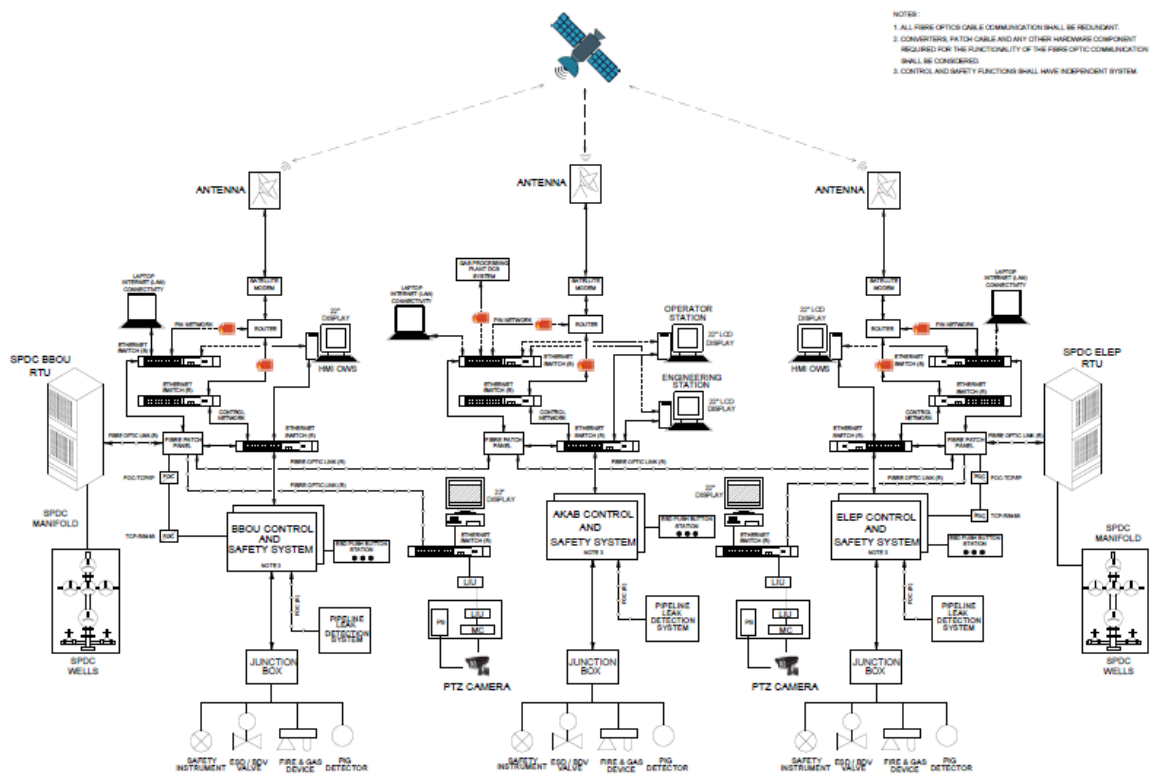
2. General

- 2.1. Vendor shall study the requirement of Satellite Communication between BBOU, ELEP and AKAB and suggest the best possible technology for the application.
- 2.2. Vendor shall be responsible for the link study and suggest the clear communication links.
- 2.3. Vendor shall take the Satellite Communication project on the lumpsum with study, design, supply, software, inspection, installation testing and certification.
- 2.4. Vendor shall assist in getting all frequency license approval from the Nigerian authority.
- 2.5. Vendor shall provide all pre-requisites for the installation of Satellite Communication System.
- 2.6. Vendor shall be responsible for the providing the detailed installation strategies of the Satellite communication systems.
- 2.7. Vendor shall follow the international codes (ITU) and standards and local regulations for the telecommunications
- 2.8. Vendor shall offer inspection and testing as per the contractual requirement.
- 2.9. Vendor shall provide all related documentation as listed in Section 5 for CLIENT review and approval.
- 2.10. Communication with the main plant on Brass Island will be achieved primarily through fibre optic cable, with VSAT system as back up. There is therefore no need for communication towers near the manifolds.

3. Technology for Satellite Communication

- 3.1. Vendor shall provide the detailed architecture for the satellite communications.
- 3.2. Satellite communication shall be used for data communication , video communication, telephony system.
- 3.3. The system shall comprise of Antenna, Low Noise Amplifier (NA), Solid State Power Amplifier (SSPA), UP-Converter, DOWN-Converter, Routers, Modem, MUX, EPABX.
- 3.4. Satellite communication shall be used for the transferring the data from unmanned platforms (BBOU and ELEP) to base station at AKAB.
- 3.5. Satellite communication shall be used in tandem with Fibre Optics Communication. In case of failure of Fibre Link Satellite Communication shall be automatically activated.
- 3.6. SCS shall carry the process variables like well data and flow line data. It also shall carry internet signals and CCTV video data.
- 3.7. Vendor shall suggest the uplink and downlink frequencies after studying the application and shall provide the equipment accordingly.
- 3.8. Cyber security aspects of the SCS shall be built-in in the SCS.
- 3.9. Vendor shall provide all hardware and software that will be required for the complete operation.
- 3.10. Vendor shall indicate the service requirement after the supply and five year CAMC shall be considered.
- 3.11. The SCS system shall communicate with the SCADA systems at BBOU and ELEP and DCS system at AKAB.
- 3.12. SCS system shall be capable of providing the link failure alarm to SCADA and DCS system.
- 3.13. Vendor shall suggest all utilities, power and pre-requisites that are required from the purchasers.

4. Typical System Architecture



5. Vendor Data Requirement Schedule (VRDS)

Sr.No	Description	With Proposals (prints)	After Purchase Order		Final documentation / As-built
			Prints	Data needed	
1	Vendor Drawing and Document Schedule		X		X
2	Cabinet/Rack /Console – GA		X		X
3	Power load & UPS requirements		X		X
4	Bill of Material		X		X
5	Functional Design Specification				X
6	Factory Testing & Acceptance procedure				X
7	Site Testing & Acceptance procedure		X		X
8	Catalogue		X		X
9	Application/System Software documentation		X		X
10	Hardware specification		X		X
11	Training Course Material		X		X
12	Installation, Operation & Maintenance Manuals		X		X
13	Two years of operational spare				
14	Mandatory Spare		X		X
15	Consumable spares, Special test tools		X		X
16	Proven Track Record				
17	Compliance Statement and deviation list (if any)				
18	Dully Filled in Logistic support Certificate				
19	Soft copy (CD/DVD/Flash Drive) of final documentation				X